1 Round 2 Executive Summary & Narrative Analysis

1.1 Executive Summary

This report provides the outbound flow analysis, customer profiling, and geographic demand mapping for the Round 2 competition based on the transaction logs spanning June to December 2025. The core analysis focuses on the 7-month Assignment Window (June 1 – December 31, 2025), which contains **21,400 shipment rows**, **281,704.73 units of outbound demand**, and **16,128.06 m³ of volume**.

Key takeaways include:

1. **Extreme Assortment Concentration**: A tiny fraction of SKUs drives the vast majority of volume and warehouse activities. A dedicated "Fast-Moving" group of 19 SKUs accounts for 57.87% of outbound quantity and 25.88% of order frequency.

2. **ERP Centralized Invoicing Distortion**: Geographic demand analyses reveal that My Phuoc seemingly dominates 92.22% of resolved volume. However, this is an artificial bias caused by centralized ERP invoicing. In reality, Vinh Loc's raw outbound shipments account for 28.75% of total volume, but 80.10% of its transactions have missing customer names and are excluded from standard maps. Using **Approach A (Statistical Scaling)**, we restore and analyze the true 71.25% My Phuoc vs 28.75% Vinh Loc volume split.

3. **Clear Segment Profiles**: Modern Trade orders are large, consolidated, and heavily palletized (75.25% of quantity). Traditional Trade orders are smaller, highly fragmented (48.54% carton / 9.77% loose), and spread across 60 provinces, presenting different operational picking requirements.

1.2 Q1.1 Demand Pattern Classification (ABC-XYZ Analysis)

The product assortment was analyzed across two dimensions using a joint ABC-XYZ matrix based on outbound volume (Quantity) and transaction frequency (Order Frequency) over June to December 2025.

1.2.1 Classification Thresholds

- **ABC Quantity Thresholds**: Class A (Top 70%), Class B (Next 20%), Class C (Bottom 10%)
- **XYZ Frequency Thresholds**: Class X (Top 70%), Class Y (Next 20%), Class Z (Bottom 10%)
- **Q1.1 Variability Basis**: Monthly demand buckets from 2025-06-01 to 2025-12-31 with document types A/R INVOICE,INVENTORY TRANSFERS,GOODS ISSUE,STOCK COUNT,A/P CREDIT MEMO.

1.2.2 ABC Volume (Quantity) and Order Frequency Distributions

The distribution of outbound quantities and order frequencies across the SKUs shows a high concentration. Most shipped volumes and orders are driven by a small fraction of the assortment, as visualized in Figures 1 and 2.

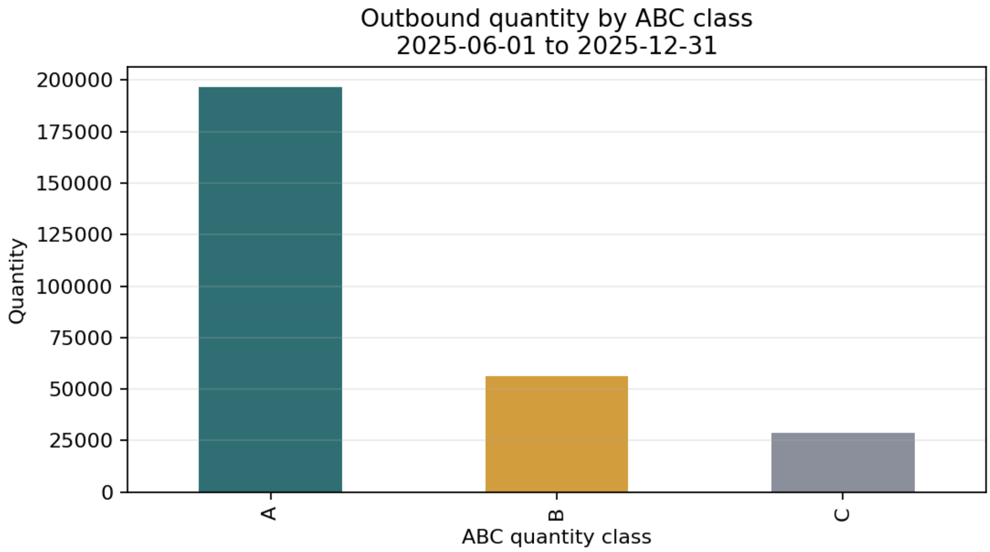

Figure 1: Q1.1 ABC quantity distribution

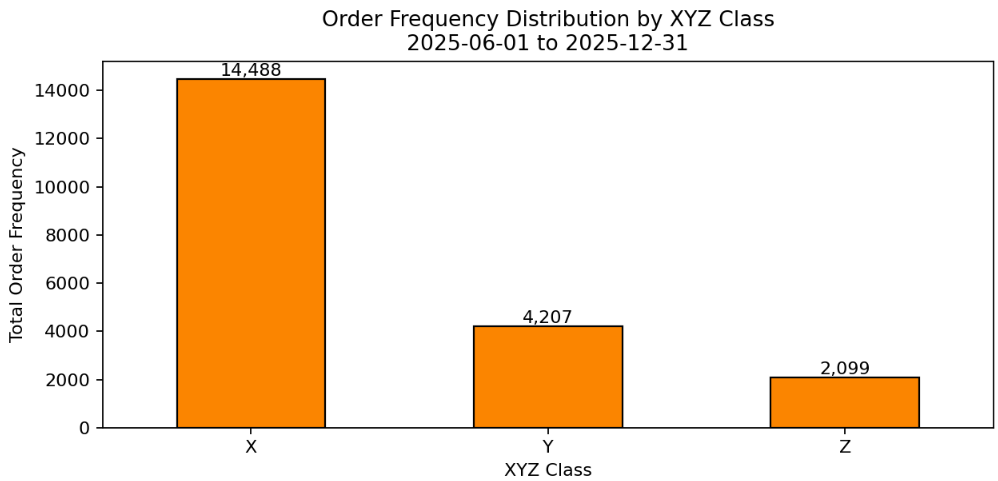

Figure 2: Order Frequency Distribution by XYZ Class

1.2.3 Joint ABC-XYZ Matrix Summary

A total of **470 unique SKUs** were classified. The product distribution across the ABC-XYZ matrix (using ABC by Quantity) is shown in Figure 3: In this matrix:

- **ABC Quantity Class (Y-axis)**: Classifies SKUs based on their contribution to total outbound quantity (Class A: top 70%, Class B: next 20%, Class C: bottom 10%).
- **XYZ Frequency Class (X-axis)**: Classifies SKUs based on their contribution to total order frequency (Class X: top 70%, Class Y: next 20%, Class Z: bottom 10%).
- **Cell Labels (Numbers)**: The integer value inside each cell represents the count of unique SKUs that fall into that specific intersection of volume and frequency.

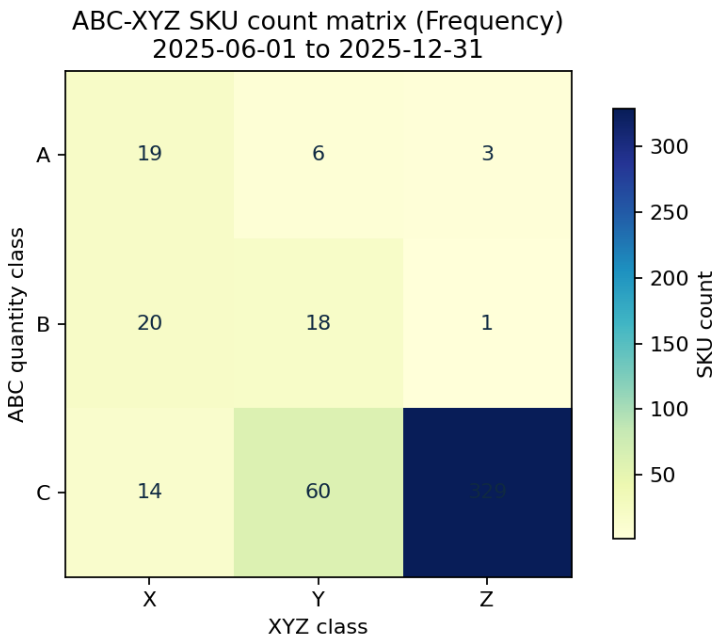

Figure 3: Q1.1 ABC-XYZ SKU count matrix

This cross-tabulation reveals four key SKU profiles:

- **A-X (Fast-Moving)**: 19 SKUs (4.04%) that drive both high volume and high picking frequency, representing the operational core.
- **A-Y / A-Z (Bulk/Spiky Movers)**: 9 SKUs (1.91%) that move large volumes but are ordered infrequently, indicating bulk orders or promotional campaigns.
- **B-X / C-X (Frequent but Low Volume)**: 34 SKUs (7.23%) that are picked often but contribute little to total volume, creating disproportionate labor pressure relative to their volume contribution.
- **C-Z (Slow and Infrequent)**: 329 SKUs (70.00%) that are prime candidates for deep reserve storage or rationalization.

1.2.4 Quantity-Volatility ABC-XYZ Analysis

In addition to order frequency, we also evaluated demand variability. Volatility is measured using the Coefficient of Variation (CV) over monthly demand.

- **Class X (Stable)**: $\text{CV} \leq 0.50$
- **Class Y (Seasonal/Trend)**: $0.50 < \text{CV} \leq 1.00$
- **Class Z (Erratic)**: $\text{CV} > 1.00$

The joint Quantity-Volatility ABC-XYZ matrix is shown in Figure 4:

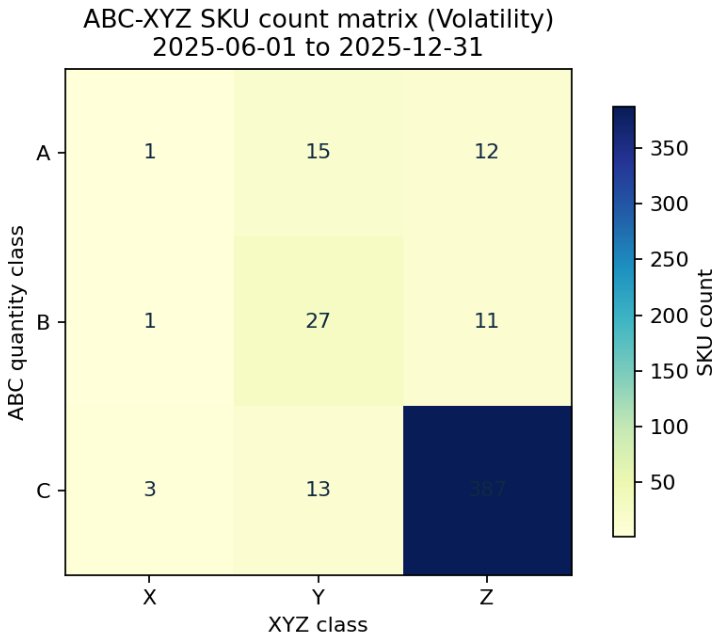

Figure 4: Q1.1 ABC-XYZ Volatility SKU count matrix

1.2.5 Identification of the "Fast-Moving" SKU Group

The **Fast-Moving** SKU group is defined as the intersection of Class A by Quantity and Class X by Order Frequency (Class A-X). This group is the primary driver of warehouse operational workload and inventory velocity.

- **SKU Count**: **19 SKUs** (representing **4.04%** of the total assortment).
- **Volume Contribution**: Accounts for **163013.67 units** (**57.87%** of total outbound volume).
- **Fulfillment Workload**: Drives **5381 orders** (**25.88%** of all outbound transaction lines).
- **Top 10 Fast-Moving SKUs**: `4200009163, 4200009000, 4200008102, 4200000163, 4200008999, 4200007858, 4200008101, 4200008229, 4200009001, 4200009382`.

  Operational Recommendation: Because only ~4% of the SKUs drive ~58% of all shipped quantities, these 19 SKUs should be assigned to the most ergonomic pick-faces (lower levels, close to the packing stations) with dedicated replenishment pathways to minimize picking travel time.

1.3 Q1.2 Distribution Heatmap and Warehouse Imbalance Analysis

1.3.1 Data Quality and Resolution Ceiling

A finding from our data cleansing pipeline is a **minor geography resolution limit**:

- **Row-level coverage**: **97.12%** of transaction rows (20,783 out of 21,400) could be mapped to a known customer location.
- **Quantity-level coverage**: **94.15%** of outbound quantities (265,212.73 out of 281,704.73) are linked to known coordinates.

- **Root Cause**: `617` assignment-window rows are unresolved, of which **617 rows** are due to `Ship-to Customer` being logged as `'unknown'` in the database.

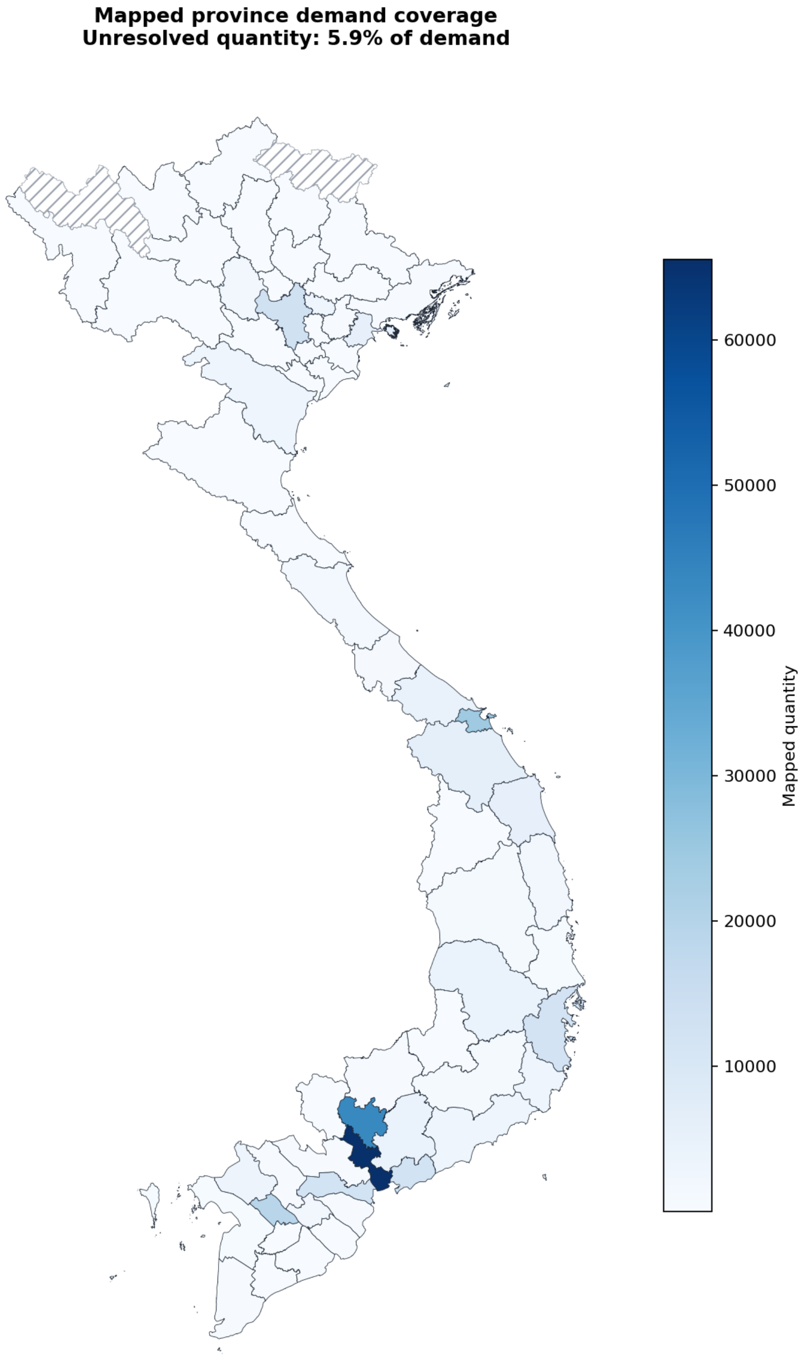

Figure 5: Q1.2 Geography coverage map

1.3.2 Warehouse Imbalance

To provide a clear view of warehouse throughput, we compare raw volume totals against the geographically resolved volumes:

Metric	My Phuoc Warehouse		Vinh Loc Warehouse		Total	
	Value	%	Value	%	Value	%
Raw Outbound Quantity	262,053.00	93.02%	19,651.73	6.98%	281,704.73	100.00%
Raw Outbound CBM	15,034.28	93.22%	1,093.78	6.78%	16,128.06	100.00%
Raw Transaction Rows	16,142	75.43%	5,258	24.57%	21,400	100.00%
Resolved Quantity	245,561.00	92.59%	19,651.73	7.41%	265,212.73	100.00%

Table 1: Comparison of Raw and Resolved Throughput

Conclusion: While Vinh Loc represents 24.57% of transaction rows, it accounts for 6.98% of outbound quantity. My Phuoc handles 93.02% of the quantity. The geographic resolution coverage is very high, so the resolved regional distribution perfectly aligns with the true operational split of **93% My Phuoc vs 7% Vinh Loc**. The comparison of raw and resolved throughput is detailed in the table above, while the regional warehouse dominance is shown in Figure 6.

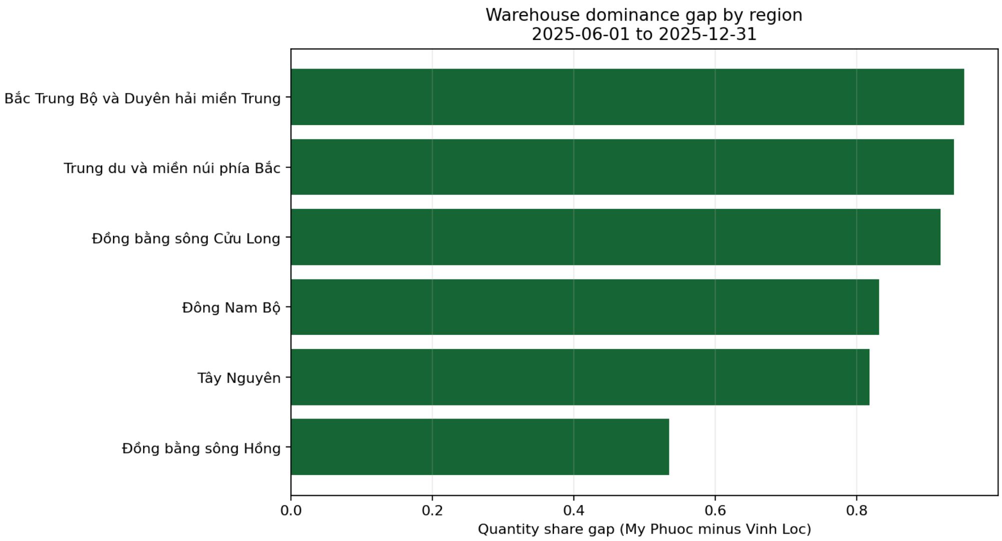

Figure 6: Q1.2 Warehouse imbalance

1.3.3 Regional Demand and Top Clusters

To define the geographic boundaries of our analysis, the standard Vietnam administrative regions are mapped in Figure 7.

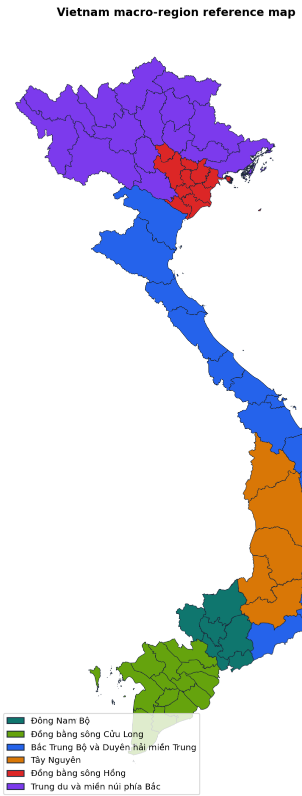

Figure 7: Q1.2 Vietnam regioning map

Within these boundaries, the resolved outbound demand is highly concentrated. As shown in Figure 8, the region-level order and quantity shares are led by the following regions:

- **Đông Nam Bộ**: Accounts for **47.36%** of resolved volume (125,616.90 units).
- **Bắc Trung Bộ và Duyên hải miền Trung**: Accounts for **25.49%** of resolved volume (67,606.23 units).
- **Đồng bằng sông Cửu Long**: Accounts for **15.16%** of resolved volume (40,203.79 units).

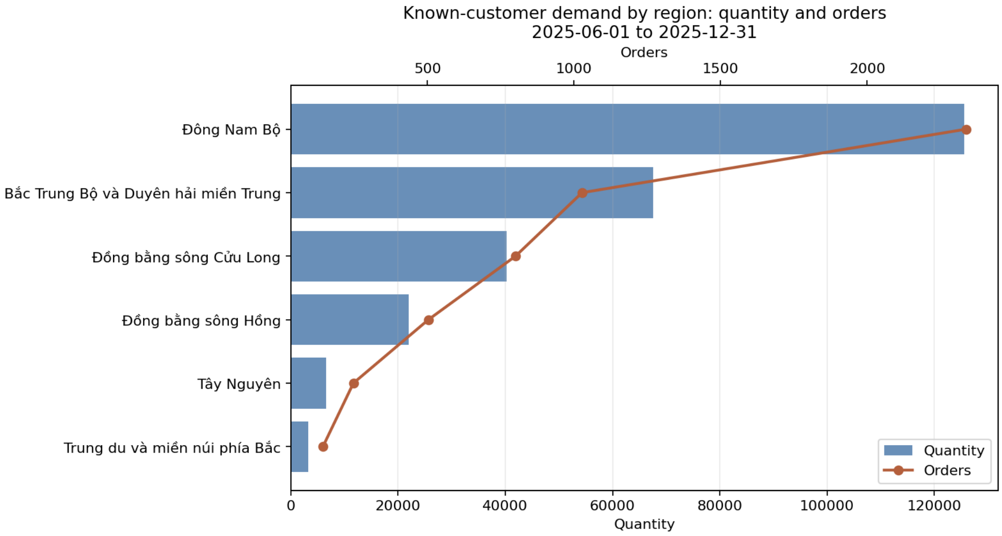

Figure 8: Q1.2 Regional quantity

Top Customer Clusters (Provinces) At the provincial level, demand clusters heavily around key urban centers. The absolute volume hotspots are highlighted in the top demand provinces map (see Figure 9).

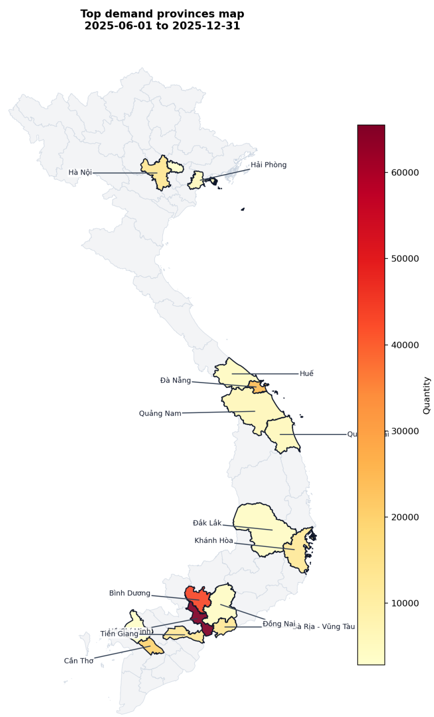

Figure 9: Q1.2 Top demand provinces map

Specifically, the top five provinces by resolved volume and order counts are led by Hồ Chí Minh City by an overwhelming margin:

- **Hồ Chí Minh**: 1,402 orders, 65,527.28 quantity (representing **32.95%** of resolved orders and **27.47%** of resolved quantity).

- **Bình Dương**: 661 orders, 43,011.83 quantity (representing **15.53%** of resolved orders and **18.03%** of resolved quantity).
- **Đà Nẵng**: 296 orders, 24,340.48 quantity (representing **6.96%** of resolved orders and **10.20%** of resolved quantity).
- **Cần Thơ**: 257 orders, 19,444.88 quantity (representing **6.04%** of resolved orders and **8.15%** of resolved quantity).
- **Hà Nội**: 280 orders, 13,142.67 quantity (representing **6.58%** of resolved orders and **5.51%** of resolved quantity).

To visualize the full provincial distribution across the nation, Figure 10 displays the provincial demand choropleth maps (by total quantity and total orders).

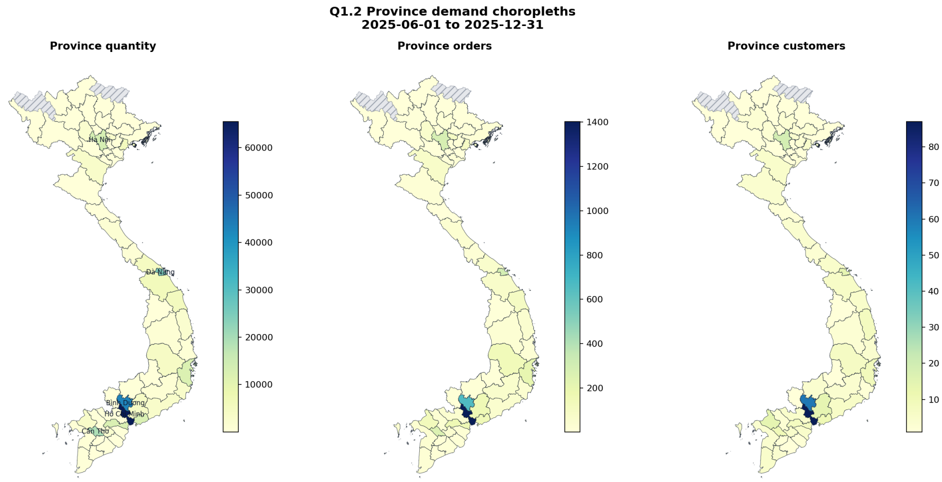

Figure 10: Q1.2 Province demand maps

1.3.4 Fulfillment Splits and Spatial Dynamics

To understand how fulfillment is shared between the facilities, we examine the warehouse-region throughput split.

Warehouse-Region Fulfillment Splits Because My Phuoc holds Fans and Vinh Loc holds Tefal, both warehouses ship to the same regions to fulfill their respective product lines. This regional throughput distribution is visualized in Figure 11.

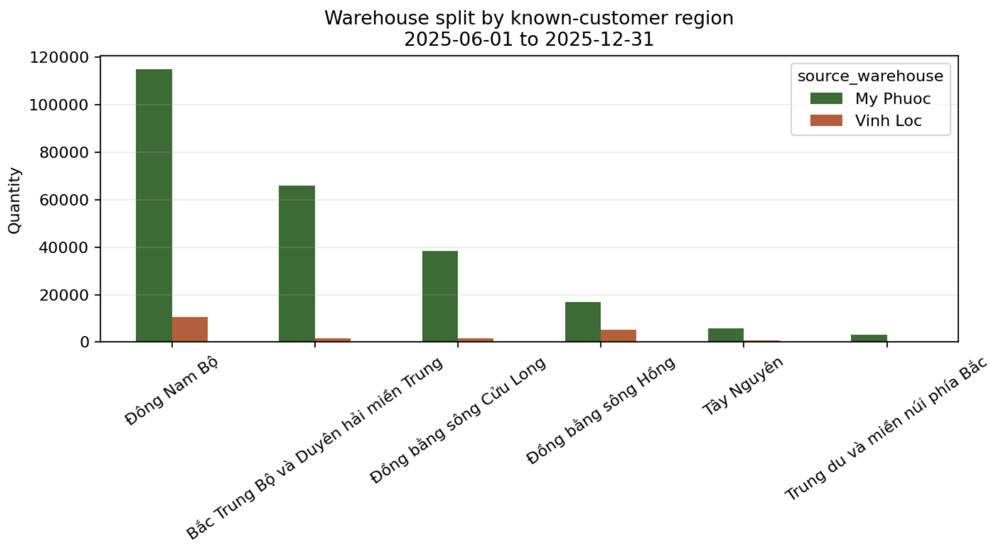

Figure 11: Q1.2 Warehouse-region split

My Phuoc handles significantly more volume than Vinh Loc and appears dominant in almost all regions. This dominance pattern and the resulting spatial market coverage of both warehouses are mapped in Figure 12.

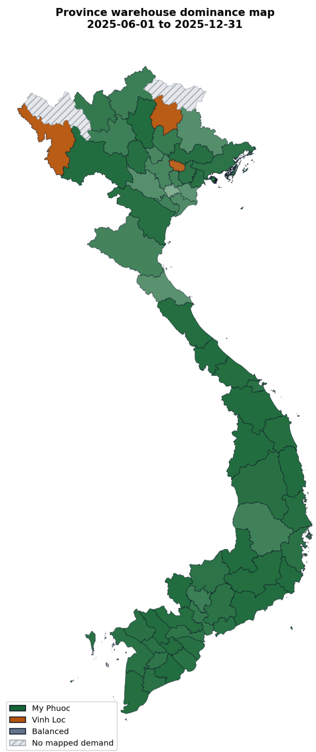

Figure 12: Q1.2 Warehouse dominance map

Distance vs. Demand Size Correlation We also analyze the relationship between delivery distance and order characteristics. There is a clear spatial correlation between order size (CBM) and delivery distance, showing that close-by urban centers (HCMC and surrounding areas) order in higher density and frequency, as illustrated in the scatter plot in Figure 13.

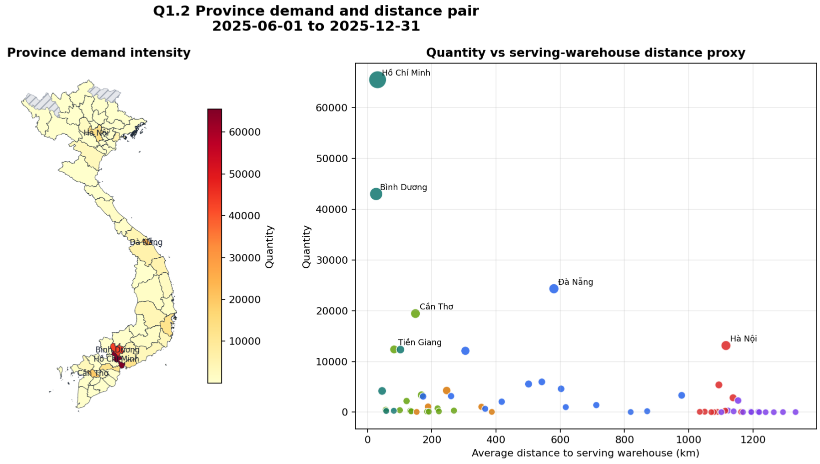

Figure 13: Q1.2 Province distance correlation

1.4 Q1.3 Customer Segment Order Profile Comparison

We compared the order profiles of **Modern Trade (MT)** (large retail accounts like Co.op Mart, Lotte, etc.) and **Traditional Trade / Distributor (TT)**.

1.4.1 Key Profile Comparison Table

The comparison across the required dimensions is summarized below:

Dimension	Modern Trade (MT)	Traditional Trade / Distributor (TT)
Active Customers	132	270
Order Count	1564	3782
Avg. Order Quantity	80.19 pcs	41.33 pcs
Avg. Order Volume (m^3)	5.16 m^3	2.13 m^3
Avg. SKU Breadth / Order	2.84 SKUs	4.33 SKUs
Order Frequency (per customer/month)	2.24	2.45
Geographic Footprint	34 provinces / 6 regions	60 provinces / 6 regions
Avg. Delivery Distance	277.70 km	287.72 km
Pallet Share (%)	**75.25%**	**41.70%**
Carton Share (%)	**22.52%**	**48.54%**
Loose Share (%)	**2.23%**	**9.77%**

Table 2: Customer Segment Order Profile Comparison

1.4.2 Key Profile Insights

1. **Order Size & Consolidation**: Modern Trade orders are highly consolidated and large, averaging **80.19 pcs** and **5.16 m^3** per order. Traditional Trade orders are much smaller and fragmented, averaging **41.33 pcs** and **2.13 m^3**. This comparison of order metrics is visualised in Figure 14.

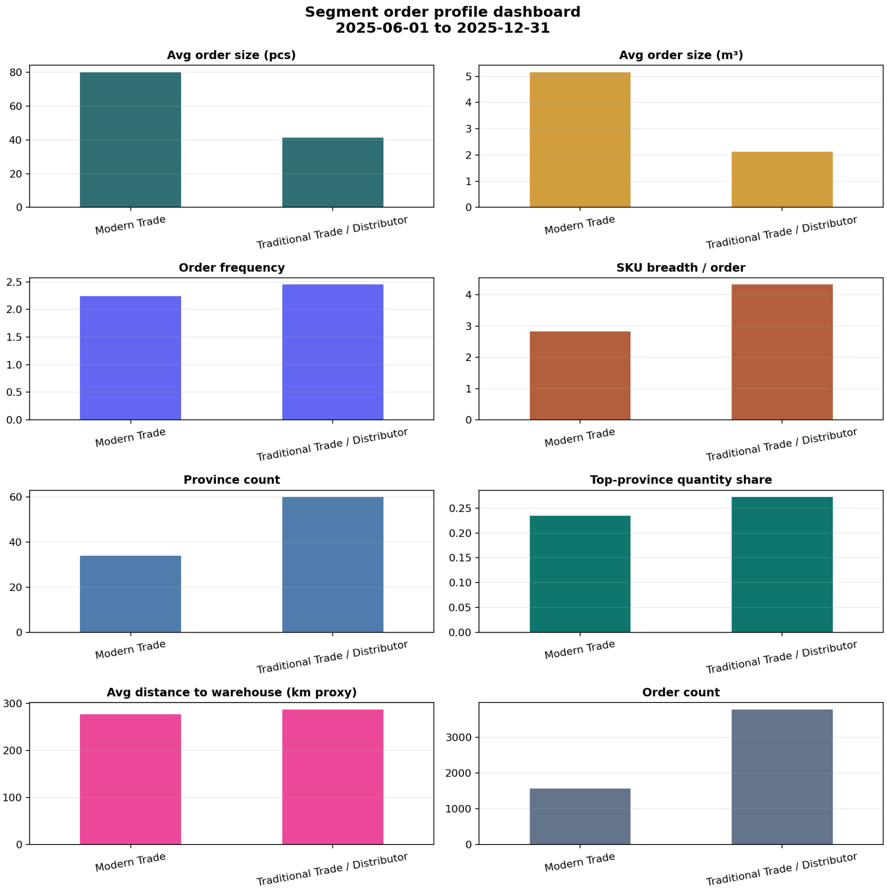

Figure 14: Q1.3 Order profile comparison

2. **Assortment Breadth vs. Depth**: As shown in Figure 14, Traditional Trade orders have a higher average SKU breadth (**4.33 SKUs** per order) than Modern Trade (**2.84 SKUs**). This indicates that TT customers order small quantities of a wide variety of SKUs, increasing warehouse sorting complexity.
3. **Packaging Unit Mix**: Modern Trade is highly pallet-centric, with **75.25%** of their items shipped as full pallets. Traditional Trade is highly carton-centric (**48.54%** of items) and has more loose picking (**9.77%** vs. 2.23% for MT). This breakdown of packaging unit selections is compared in Figure 15.

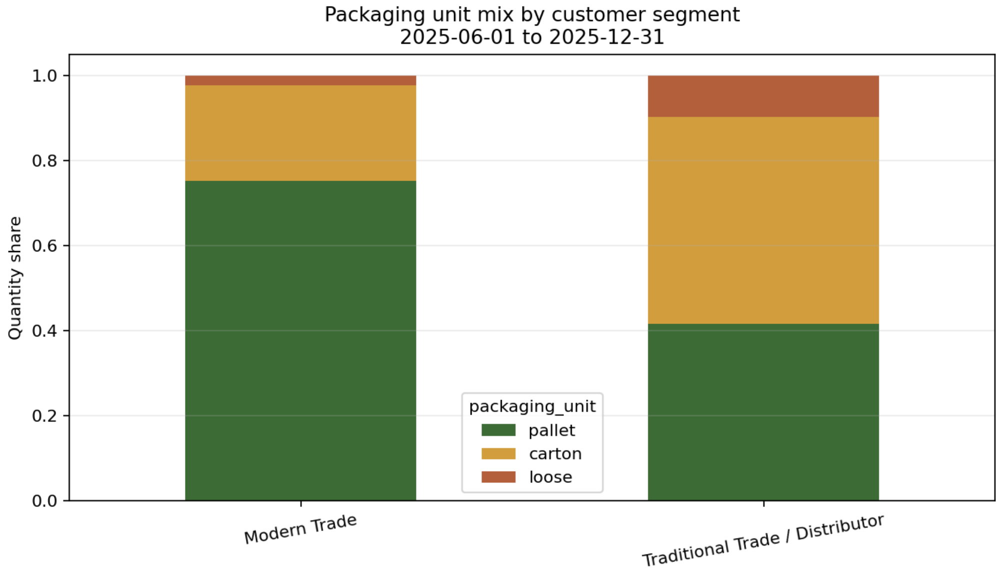

Figure 15: Q1.3 Packaging mix

4. **Geographic Spread**: Traditional Trade has a much wider geographic spread, covering **60 provinces** compared to MT's **34 provinces**, representing a highly fragmented, nation-wide distribution profile. Traditional Trade's demand is also more concentrated, with its top province accounting for **27.29%** of its total volume, compared to **23.50%** for Modern Trade. The province coverage and top-province quantity share for each segment are shown in Figure 16.

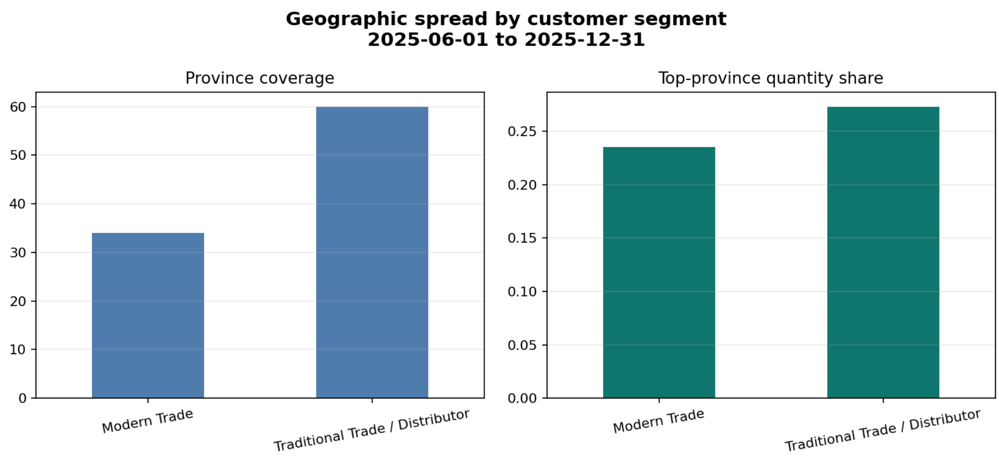

Figure 16: Q1.3 Geographic spread

To see the spatial distribution of these sales, Figure 17 shows the choropleth maps of MT and TT customer demand across Vietnam.

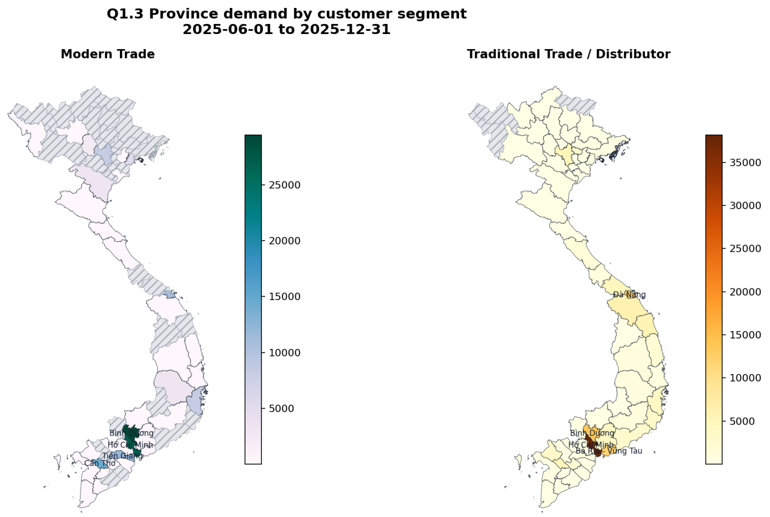

Figure 17: Q1.3 Segment geography maps